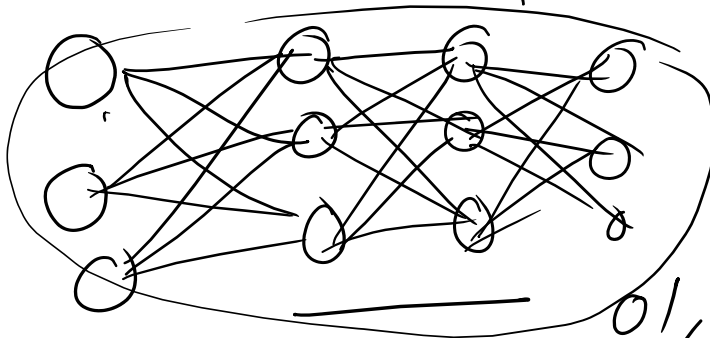


DL → Keras with tensorflow

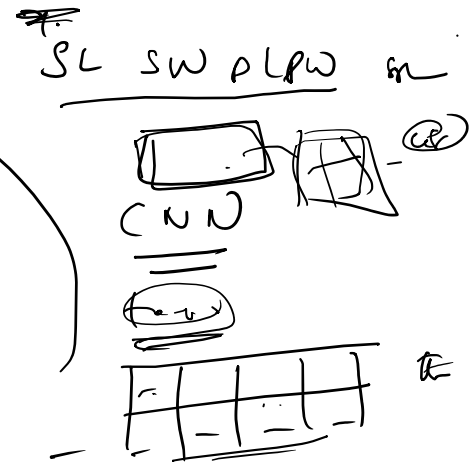
- ① Sequential
- ② functional
- ③ Model subclassing



ANN

IL HL
↓
CNN

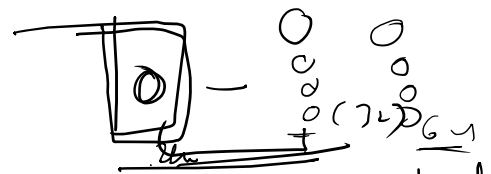
conv maxpooling flatten
+ kernel (3,3)



mnist data sample

28 x 28 [0-9]

model = Sequential()



model.add(^{Conv2D}Dense(32, (3,3) activation='relu'
input_shape=(4,28,28))

model.add(MaxPooling2D(2,2)),

model.add(conv2D(64, (3,3) activation='relu'))

```

model.add(Conv2D(64, (3,3), activation = 'relu'))
model.add(MaxPooling2D(2,2)),
model.add(Flatten()),
model.add(Dense(64, activation = 'relu'))
model.add(Dense(10, activation = 'softmax'))

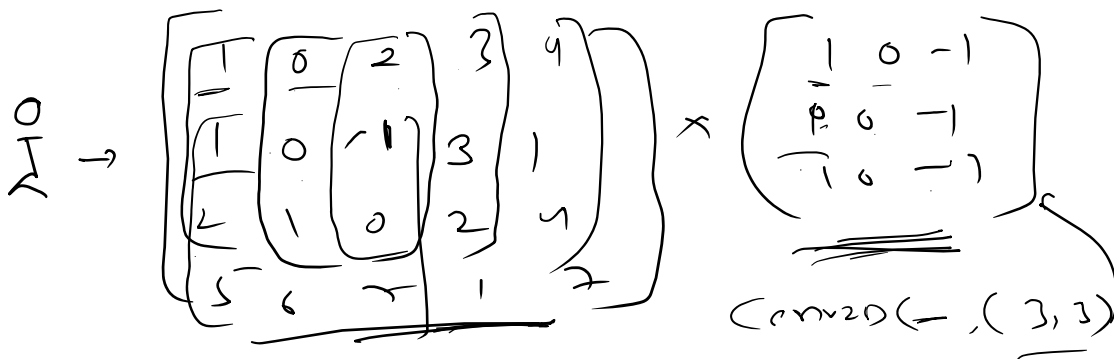
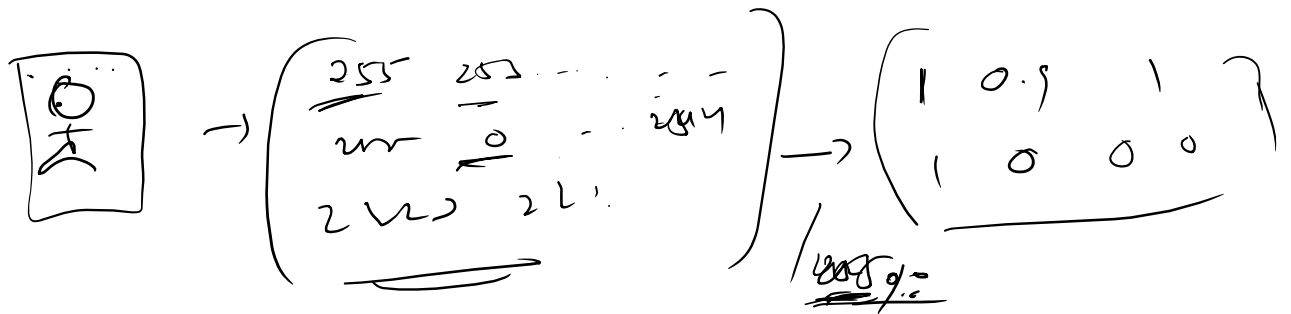
```

Sequential

0 1 2 3 4 5 6 7 8 9

train = 60,000

test = 10,000



$$= 1 + 0 + (-2) + 1 + 0 + (-1) + 2 + 0 + 0(-1) = 0$$

$$= 0 + 0 - 3 + 0 + 0 - 3 + 1 + 0 - 2 = -7$$

$$\begin{bmatrix} 0 & -7 & 3 \\ 2 & 1 & 6 \end{bmatrix}$$

28x28 → (3,3)

output

0 1 2 3 4 5 6 7 8 9

$$\begin{pmatrix}
 1 & 2 & 3 & 4 & 5 \\
 0 & 1 & 2 & 1 & 5 \\
 6 & 1 & 2 & 1 & 1 \\
 1 & -1 & 2 & 3 & 5 \\
 1 & 0 & 1 & 2 & 3
 \end{pmatrix}
 =
 \begin{pmatrix}
 2 & 3 & 4 & 5 \\
 6 & 2 & 2 & 5 \\
 6 & 2 & 3 & 5 \\
 1 & 2 & 3 & 5
 \end{pmatrix}$$

4x4

5 X 5